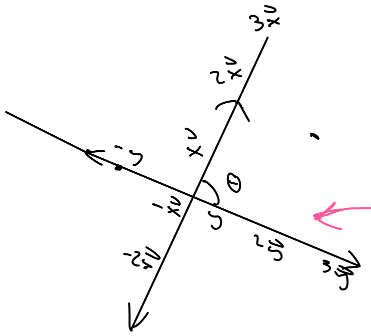
Let's impose geometry on $\vec{x}$ and $\vec{y}$.

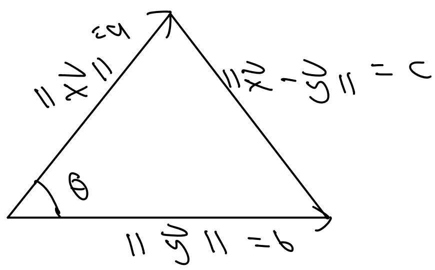
- imagine $\vec{x}$ and $\vec{y}$ "sit" in $\mathbb{R}^n$ space on a "plane" $= \{a\vec{x}+b\vec{y} : a,b \in \mathbb{R}\ \ \vec{x} \nparallel \vec{y}\}$



- not an actual plane; real planes are in $\mathbb{R}^3$
- HOWEVER, we can imagine this to impose geometry

Let's look at the "plane."

- if we add $\vec{y}$ to $\vec{x}$, we see:



By Law of Cosines,

$$c^2 = a^2 + b^2 - 2ab\cos(\theta)$$

$$\|$$

$$\|\vec{x}-\vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|_2^2 - 2\|\vec{x}\|_2\|\vec{y}\|_2\cos\theta$$

Recall: $\|\vec{z}\|_2^2 = \vec{z}\cdot\vec{z}$

$$\|\vec{x}-\vec{y}\|_2^2 = (\vec{x}-\vec{y})\cdot(\vec{x}-\vec{y})$$

$$= \vec{x}\cdot(\vec{x}-\vec{y}) - \vec{y}\cdot(\vec{x}-\vec{y})$$

bilinearity

$$= \vec{x}\cdot\vec{x} - \vec{x}\cdot\vec{y} - \vec{y}\cdot\vec{x} + \vec{y}\cdot\vec{y}$$

$$= \|\vec{x}\|_2^2 - 2\cdot\vec{x}\cdot\vec{y} + \|\vec{y}\|_2^2$$

$-2\cdot\vec{x}\cdot\vec{y} = -2\|\vec{x}\|_2\|\vec{y}\|_2\cos\theta$??

$a=b$
$a=c$

transitive property

$$\Rightarrow \underbrace{\|\vec{x}\|_2^2}_{A} + \underbrace{\|\vec{y}\|_2^2}_{B} \underbrace{-2\|\vec{x}\|_2\|\vec{y}\|_2\cos\theta}_{C} = \underbrace{\|\vec{x}\|_2^2}_{A} \underbrace{-2\cdot\vec{x}\cdot\vec{y}}_{C} + \underbrace{\|\vec{y}\|_2^2}_{B}$$

$$\Rightarrow -2\|\vec{x}\|_2\|\vec{y}\|_2\cos\theta = -2\cdot\vec{x}\cdot\vec{y}$$

$$\Rightarrow \|\vec{x}\|_2\|\vec{y}\|_2\cos\theta = \vec{x}\cdot\vec{y} \quad \checkmark$$

Intuition

$\Rightarrow \quad \cos\theta = \dfrac{\vec{x}}{\|\vec{x}\|_2} \cdot \dfrac{\vec{y}}{\|\vec{y}\|_2}$ where $\vec{x} \neq \vec{0}$, $\vec{y} \neq \vec{0}$

Claim:

If $\vec{u} = \dfrac{\vec{x}}{\|\vec{x}\|_2}$, then $\underbrace{\|\vec{u}\|_2}_{\text{unit vector (length 1)}} = 1$

Proof. Let $\vec{x} \in \mathbb{R}^n$ w $\vec{x} \neq \vec{0}$

set $\alpha = \dfrac{1}{\|\vec{x}\|_2} > 0$ positivity

$\|\vec{x}\|_2^2 - 2 \cdot \vec{x} \cdot \vec{y} + \|\vec{y}\|_2^2$

Consider:

$$\|u\|_2 = \left\| \frac{x}{\|\vec{x}\|_2} \right\|_2$$

$$= \left\| \underbrace{\frac{1}{\|\vec{x}\|_2}} \cdot \vec{x} \right\|_2$$

$$= \|\alpha \cdot \vec{x}\|_2$$

$$= |\alpha| \cdot \|\vec{x}\|_2$$

$$= \alpha \cdot \|\vec{x}\|_2$$

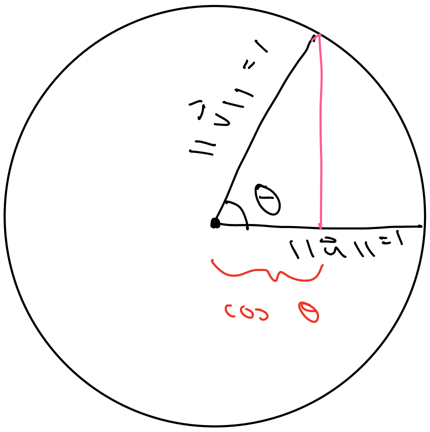
$$= \frac{1}{\|\vec{x}\|_2} \cdot \|\vec{x}\|_2$$

$\Rightarrow \quad \|\vec{u}\|_2 = 1$ ✓

$\blacksquare$

$\underbrace{\dfrac{\vec{x}}{\|\vec{x}\|}}_{\vec{u}} \cdot \underbrace{\dfrac{\vec{y}}{\|\vec{y}\|}}_{\vec{v}} = \cos\theta$ where θ angle between $\vec{x}$ and $\vec{y}$

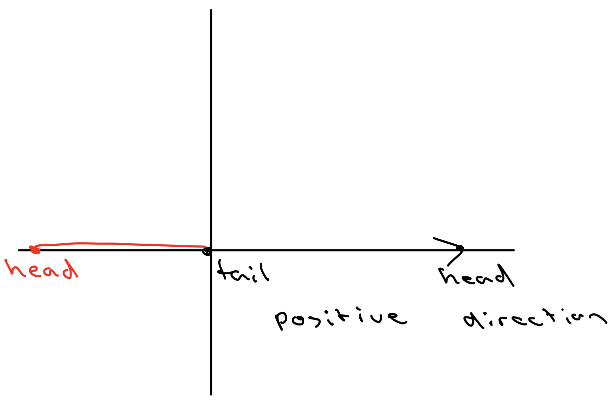
unit vectors



amount of $\vec{v}$
in direction of $\vec{u}$

Direction = span (set of all scalar multiples)

Direction ≠ Orientation?



read left to right

head → tail but on same span
∴ opposite orientation

14 min